

MUTHUKUMAR RAMALINGAM

Lead Software Engineer · .NET Distributed Systems · Cloud & AI

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PROFESSIONAL SUMMARY

Lead Software Engineer with 12+ years building cloud-native .NET microservices on AWS and Azure. Delivered 99.9% uptime systems using Kafka event streaming, Saga Orchestration, and Polly resilience patterns — cutting high-volume workflow processing time by ~35%. Drives team velocity 20–25% through GitHub Copilot (GPT-4o) and Cursor IDE (Claude Opus 4); led AI invoice automation reducing document processing from 8 min to 40 sec.

CORE TECHNICAL SKILLS

Languages	C# (.NET Core / .NET 9), JavaScript / TypeScript, Python, SQL
Architecture	Microservices, Saga Orchestration, DDD, Clean Architecture, CQRS, Event Sourcing, SOLID
Cloud & Messaging	AWS (SQS, SNS, Lambda, S3, EC2), Azure (AKS, Document Intelligence, OpenAI), Kafka (Confluent), Docker, Kubernetes
Front-End	React JS, Material UI, Responsive Design, REST/GraphQL API Integration
Platform Eng.	Redis Cache, Polly (Circuit Breaker, Retry), Datadog APM, OpenTelemetry, Prometheus, OWASP / JWT / IAM
AI & LLM	Azure OpenAI (GPT-4o), Prompt Engineering, Structured Output, RAG pipelines, GitHub Copilot, Cursor IDE (Claude Opus 4)
Testing & Process	TDD, xUnit, Moq, FluentAssertions, Contract Testing, Agile Scrum, ADRs, Code Review & Mentorship

PROFESSIONAL EXPERIENCE

Lead Software Engineer | Trimble, Chennai *April 2021 – Present*

- Architected multi-tenant Connected Maintenance platform with Saga Orchestrator microservices across 10+ service domains — zero data inconsistency incidents since launch.
- Sustained 99.9% uptime applying Polly Circuit Breaker and Exponential Backoff; migrated to Confluent Kafka cutting high-volume workflow processing time by ~35% and doubling throughput under peak load.
- Cut MTTD from ~45 min to ~25 min by integrating OpenTelemetry with Datadog APM; reduced API response times 12–15% via Redis Distributed Cache on critical read paths.
- Standardised AI-assisted development using GitHub Copilot (GPT-4o) and Cursor IDE (Claude Opus 4): authored shared .cursorrules enforcing DDD naming and Clean Architecture conventions, built reusable Cursor Skills for Kafka consumer and Polly policy scaffolding, and created sprint-level AI Plans to generate test skeletons from acceptance criteria — lifting TDD coverage from 62% to 85% and cutting PR cycle time by 20%.
- Proxy Product Owner: led Agile Scrum ceremonies, stakeholder communication, and mentored 6 engineers with 2 promoted to senior roles.

Senior Software Engineer | Trimble, Chennai *December 2018 – March 2021*

- Reduced peak CPU spikes ~40% on the Sightfusion crack-detection pipeline by decoupling processing onto AWS SQS with independent horizontal auto-scaling.
- Improved ETL throughput ~30% by provisioning Lambda concurrency and restructuring sequential steps into parallel fan-out patterns; ran A/B tests on UI variants to drive conversion improvements.

Programmer Analyst | Boeing, Bangalore *August 2017 – December 2018*

- Built BMAPS aircraft production scheduling system with R-Tree spatial indexing and Precedence Network algorithms — cut Gantt chart computation time by 50%.

- ▶ Optimised mission-critical assembly databases via SQL Profiler and execution plan analysis, cutting critical query runtimes by 45%.

Software Developer | **RGS, Chennai** July 2014 – July 2017

- ▶ Delivered BIM 3D structural plan generation from point-vector inputs for 5+ enterprise construction clients; cut memory footprint ~25% via in-memory geometry cache with LRU eviction.
- ▶ Architected WCF-based parallel workload management supporting 50 concurrent users with no performance degradation.

FEATURED ENGINEERING WORK

C# Insights SDK — Cross-Platform Analytics & Telemetry Library

Trimble · Senior Engineer · .NET Core, AWS S3, SQLite, REST

- ▶ Shipped a cross-platform C# SDK adopted by 3 product teams — instrumented NPS touchpoints and feature engagement signals, enabling product to correlate promoter scores with workflow events and cut integration effort from 2 weeks to 3 days.
- ▶ Published as an internal NuGet package with 90%+ xUnit/Moq coverage, retry-safe event flushing, SQLite offline buffering, and pre-built Datadog dashboards.

AI Invoice Import — Intelligent Document Extraction & Parts Enrichment

Trimble · Lead Engineer · Azure Document Intelligence, GPT-4o, .NET Core, React JS

- ▶ Extracted structured data from supplier PDFs and scanned images using Azure Document Intelligence across 50+ varied layouts; an LLM layer (GPT-4o via Azure OpenAI) normalised raw JSON to a canonical parts schema, resolving field inconsistencies across supplier templates.
- ▶ LLM web-search enrichment resolved missing part numbers by description and supplier context — reducing unmatched line items by ~65% and achieving ~78% straight-through processing with no human intervention.
- ▶ Cut average invoice processing time from ~8 minutes to under 40 seconds; React JS metrics dashboard tracked confidence scores, match rates, and override rates — showing 72% reduction in data-entry errors.

Trimble Drop — Cloud File Sharing Service

Trimble · Senior Engineer · .NET Core, AWS S3, CloudFront, IAM

- ▶ Designed and shipped a high-capacity file-sharing service using multipart S3 uploads, automated lifecycle policies, and pre-signed URL access control — supporting files up to 10 GB with sub-5s upload initiation and zero credential exposure.
- ▶ Integrated CloudFront CDN for download acceleration and IAM-scoped policies for per-user access isolation; adopted across 4 internal teams for large asset distribution.

Developer Productivity Tooling — AI-Assisted Scaffolding & CLI Utilities

Trimble · Lead Engineer · Cursor IDE (Claude Opus 4), .NET Core, Bash, GitHub Actions

- ▶ Built CLI scaffolding using Cursor IDE (Claude Opus 4) with authored .cursorrules for architecture conventions, reusable Skills for Kafka/Polly templates, and AI Plans to generate test skeletons from acceptance criteria — cutting new service bootstrap from ~3 days to under 4 hours.
- ▶ Adopted by 5 teams within the first month; GitHub Actions pipeline runs scaffolding as part of repo initialisation, ensuring every new service starts compliant with team standards.

KEY QUANTIFIED ACHIEVEMENTS

 12–15% API response time cut — Redis caching	 99.9% uptime — Polly Circuit Breaker + Backoff
 ~40% peak CPU spike reduction — AWS SQS decoupling	 20–25% sprint throughput — GitHub Copilot + Cursor
 MTTD: ~45 min → ~25 min — Datadog + OpenTelemetry	 62% → 85% test coverage — TDD + Copilot test gen

EDUCATION

Bachelor of Engineering — Computer Science & Engineering 2011 – 2014

AVC College of Engineering, Mayiladuthurai · Anna University